

gilmore-lie-groups — formula report

3473 inline formulas (section omitted; all rendered), 1037 display equations, 38 tables, 50 unrecovered image regions.

Unresolved (10)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
gilmore-lie-groups_ EQ0006	018	—	$\begin{array}{r} I \rightarrow \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad (123) \rightarrow \left[\begin{array}{lll} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \right] \quad (321) \rightarrow \left[\begin{array}{lll} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \\ \\ (12) \rightarrow \left[\begin{array}{lll} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad (23) \rightarrow \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right] \quad (13) \rightarrow \left[\begin{array}{lll} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right] \end{array}$	(not rendered)	
gilmore-lie-groups_ EQ0113	049	—	$\begin{array}{cc} c=(a,b)=a+ib & a \in \mathbb{R}, \ b \in \mathbb{R}, \ i^2=-1 \\ (a,b) \longrightarrow \left[\begin{array}{ll} a & b \\ -b & a \end{array} \right] & i=(0,1) \longrightarrow \left[\begin{array}{ll} 0 & 1 \\ -1 & 0 \end{array} \right] \end{array}$	(not rendered)	
gilmore-lie-groups_ EQ0369	115	—	$\begin{array}{ll} \operatorname{EXP} \left[\begin{array}{cc} a & b+c \\ b-c & -a \end{array} \right] & \\ = \left[\begin{array}{cc} \cosh r + a \sinh r/r & (b+c) \sinh r/r \\ (b-c) \sinh r/r & \cosh r - a \sinh r/r \end{array} \right] & r^2 = a^2 + b^2 - c^2 > 0 \\ = \left[\begin{array}{cc} 1+a & b+c \\ b-c & 1-a \end{array} \right] & a^2 + b^2 - c^2 = 0 \\ = \left[\begin{array}{cc} \cos r + a \sin r/r & (b+c) \sin r/r \\ (b-c) \sin r/r & \cos r - a \sin r/r \end{array} \right] & -r^2 = a^2 + b^2 - c^2 < 0 \end{array}$	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
gilmore-lie-groups_ E00381	119	—	$\begin{array}{c} \frac{i}{2} \left[\begin{array}{cc} 0 & ib_1 + b_2 \\ ib_1 - b_2 & 0 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array} \right] \\ \text{EXP} \downarrow \qquad \qquad \downarrow \qquad \downarrow \text{EXP} \\ [SU(1, 1)/U(1)] \quad \times \left[\begin{array}{cc} e^{+ib_3/2} & 0 \\ 0 & e^{-ib_3/2} \end{array} \right] \end{array}$	(not rendered)	
gilmore-lie-groups_ E00584	188	—	$\begin{array}{c} \left. \left. \left. \begin{array}{c} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & \ddots & \\ & & & 1 \\ & & & & 1 \\ & & & & & -1 \\ & & & & & & 1 \\ & & & & & & & -1 \\ & & & & & & & & \ddots \end{array} \right. \right. \right. \\ \begin{array}{c} H_1 \\ H_2 \\ H_3 \\ \vdots \\ H_n \\ \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\beta + E_{-\beta}) \\ \frac{1}{\sqrt{2}}(E_\beta - E_{-\beta}) \\ \vdots \end{array} \end{array}$	(not rendered)	
gilmore-lie-groups_ E00590	190	—	$\begin{array}{c} \frac{s}{u}(2) = \frac{u}{s}(1) + \frac{s}{u}(2) - \frac{u}{s}(1) \\ \frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{i}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right] \end{array}$	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
gilmore-lie-groups_ E06601	193	—	$\begin{array}{r} \mathfrak{su}(2)=\frac{1}{2}\left[\begin{array}{cc} i a_3 & i a_1+a_2 \\ i a_1-a_2 & -i a_3 \end{array} \right] \rightarrow \frac{1}{2}\left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{i}{2}\left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right] \\ \downarrow \quad \quad \quad \mathfrak{p} \rightarrow i\mathfrak{p} \downarrow \\ \mathfrak{sl}(2;\mathbb{R})=\frac{1}{2}\left[\begin{array}{cc} a_3 & a_1+a_2 \\ a_1-a_2 & -a_3 \end{array} \right] \leftarrow \frac{1}{2}\left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{1}{2}\left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right] \end{array}$	(not rendered)	$\mathfrak{su}(2)=\frac{1}{2}\left[\begin{array}{cc} i a_3 & i a_1+a_2 \\ i a_1-a_2 & -i a_3 \end{array} \right] \rightarrow \frac{1}{2}\left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{i}{2}\left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right]$ $\downarrow \quad \quad \quad \mathfrak{p} \rightarrow i\mathfrak{p} \downarrow$ $\mathfrak{sl}(2;\mathbb{R})=\frac{1}{2}\left[\begin{array}{cc} a_3 & a_1+a_2 \\ a_1-a_2 & -a_3 \end{array} \right] \leftarrow \frac{1}{2}\left[\begin{array}{cc} 0 & +a_2 \\ -a_2 & 0 \end{array} \right] + \frac{1}{2}\left[\begin{array}{cc} a_3 & a_1 \\ a_1 & -a_3 \end{array} \right]$
gilmore-lie-groups_ E06741	232	—	$\begin{array}{ccc} \lim \epsilon \rightarrow \infty & \epsilon J_{\pm} \rightarrow h_{\pm} & 2j\epsilon^2 \rightarrow 1 \\ j \rightarrow +\infty, m \rightarrow -\infty & J_3 + \frac{1}{2\epsilon^2} \rightarrow h_3 & \left \begin{array}{c} j \\ m \end{array} \right\rangle \rightarrow \left \begin{array}{c} \infty \\ n \end{array} \right\rangle \\ j+m=n \text{ (finite)} & \theta \rightarrow \frac{\pi}{2} - \sqrt{2}\epsilon x & \end{array}$	(not rendered)	$\lim \epsilon \rightarrow \infty \quad \epsilon J_{\pm} \rightarrow h_{\pm} \quad 2j\epsilon^2 \rightarrow 1$ $j \rightarrow +\infty, m \rightarrow -\infty \quad J_3 + \frac{1}{2\epsilon^2} \rightarrow h_3 \quad \left \begin{array}{c} j \\ m \end{array} \right\rangle \rightarrow \left \begin{array}{c} \infty \\ n \end{array} \right\rangle$ $j+m=n \text{ (finite)} \quad \theta \rightarrow \frac{\pi}{2} - \sqrt{2}\epsilon x$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
gilmore-lie-groups_ E08474 • C +114	145	■ 0.233	$\begin{gathered} \eta\left(\hat{n}+\frac{1}{2}\right)+R a^{\dagger 2}+L a^2+r a^{\dagger}+l a+\delta I \\ a^{\dagger 2}+L a^2+r a^{\dagger}+l a+\delta I \\ \mathfrak{d e f}=\left[\begin{array}{cccc} 0 & l & r & -2 \delta \\ 0 & \eta & 2 R & -r \\ 0 & -2 L & -\eta & l \\ 0 & 0 & 0 & 0 \end{array}\right] \\ \Re g=\left[\begin{array}{ccc ccc} 0 & -2 R & 2 L & -r & l & 0 \\ 4 L & 2 \eta & 0 & 2 l & 0 & 0 \\ -4 R & 0 & -2 \eta & 0 & -2 r & 0 \end{array}\right] \begin{array}{l} \hat{n}+\frac{1}{2} \\ a^{\dagger 2} \\ a^2 \\ a^{\dagger} \\ a \\ I \end{array} \end{gathered}$	(not rendered)	$\eta\left(\hat{n}+\frac{1}{2}\right)+R a^{\dagger 2}+L a^2+r a^{\dagger}+l a+\delta I$ $\mathfrak{d e f}=\left[\begin{array}{cccc} 0 & l & r & -2 \delta \\ 0 & \eta & 2 R & -r \\ 0 & -2 L & -\eta & l \\ 0 & 0 & 0 & 0 \end{array}\right]$ $\Re g=\left[\begin{array}{ccc ccc} 0 & -2 R & 2 L & -r & l & 0 \\ 4 L & 2 \eta & 0 & 2 l & 0 & 0 \\ -4 R & 0 & -2 \eta & 0 & -2 r & 0 \end{array}\right] \begin{array}{l} \hat{n}+\frac{1}{2} \\ a^{\dagger 2} \\ a^2 \\ a^{\dagger} \\ a \\ I \end{array}$
gilmore-lie-groups_ E08556 • C +25	174	■ 0.447	$\begin{aligned} \left\{\left[H_i, H_j\right]\right\} &=0 \\ \left\{\left[\mathbf{H}, E_{\alpha}\right]\right\} &=\alpha E_{\alpha} \\ \left\{\left[E_{\alpha}, E_{\beta}\right]\right\} &=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ &=N_{\alpha, \beta} E_{\alpha+\beta} \end{aligned}$	$\begin{aligned} \left[H_i, H_j\right] &=0 \\ \left[\mathbf{H}, E_{\alpha}\right] &=\alpha E_{\alpha} \\ \left[E_{\alpha}, E_{\beta}\right] &=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ &=N_{\alpha, \beta} E_{\alpha+\beta} \end{aligned}$	$\begin{aligned} \left[H_i, H_j\right] &=0 \\ \left[\mathbf{H}, E_{\alpha}\right] &=\alpha E_{\alpha} \\ \left[E_{\alpha}, E_{\beta}\right] &=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ &=N_{\alpha, \beta} E_{\alpha+\beta} \quad \alpha+\beta \neq 0 \text { a root } \\ &=0 \quad \alpha+\beta \text { not a root } \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
gilmore-lie-groups_ EQ0983 ●C -23	306	■0.902	$\begin{gathered} \text{ monomial } \\ y^2/x \\ y/x \\ 1 \\ x \\ y \\ xy \\ x^2y \\ xy^2 \\ xy^3 \\ xy^4 \end{gathered} \quad \text{quad} \left[\begin{array}{rrrrrr} \xi_{00} & \xi_{10} & \xi_{01} & \eta_{00} & \eta_{10} & \eta_{01} \\ 0 & 0 & -2 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & +1 & 0 & 0 \\ 0 & 0 & 0 & 0 & +2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & +1 & 0 & 0 & 0 \end{array} \right] \left[\begin{array}{c} \xi_{00} \\ \xi_{10} \\ \xi_{01} \\ \eta_{00} \\ \eta_{10} \\ \eta_{01} \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array} \right]$	$\begin{array}{l} \text{monomial} \\ y^2/x \\ y/x \\ 1 \\ x \\ y \\ xy \\ x^2y \\ xy^2 \\ xy^3 \\ xy^4 \end{array} \left[\begin{array}{rrrrrr} \xi_{00} & \xi_{10} & \xi_{01} & \eta_{00} & \eta_{10} & \eta_{01} \\ 0 & 0 & -2 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & +1 & 0 & 0 \\ 0 & 0 & 0 & 0 & +2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & +1 & 0 & 0 & 0 \end{array} \right] \left[\begin{array}{c} \xi_{00} \\ \xi_{10} \\ \xi_{01} \\ \eta_{00} \\ \eta_{10} \\ \eta_{01} \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array} \right]$	
gilmore-lie-groups_ EQ0024 ●C +21	024	■0.222	$\left.\left.\left(r_1-r_2\right)\right)\right)\backslash\right\}\rightarrow I+\left(r_1-r_2\right),$	$\left.\left.\left(r_1-r_2\right)\right)\right\}\xrightarrow{I}+\left(r_1-r_2\right),$	$\left.\left.\left(r_1-r_2\right)\right)\right\}\xrightarrow{(12)}\left(r_2-r_1\right)=-\left(r_1-r_2\right)$
gilmore-lie-groups_ EQ0490 ●C +20	150	■0.946	$\mathfrak{g}=\underbrace{\left(\hat{n}+\frac{1}{2}, a^{\dagger 2}, a^2\right)}_{\text{su}(1,1)}, a^{\dagger 2}, a^2\right)\}_{\operatorname{su}(1,1)}+\underbrace{\left(a^{\dagger}, a, I\right)}_{\text{nilpotent }}$	$\mathfrak{g}=\underbrace{\left(\hat{n}+\frac{1}{2}, a^{\dagger 2}, a^2\right)}_{\text{su}(1,1)}+\underbrace{\left(a^{\dagger}, a, I\right)}_{\text{nilpotent }}$	$\mathfrak{g}=\underbrace{\left(\hat{n}+\frac{1}{2}, a^{\dagger 2}, a^2\right)}_{\text{su}(1,1)}+\underbrace{\left(a^{\dagger}, a, I\right)}_{\text{nilpotent invariant subalgebra}}$

Conf. MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

— no value

Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

The remaining 274 flagged rows are stated as a count.

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 43 are component (C), 231 weak (W). By MathPix confidence: 179 at ≥ 0.9 , 45 in 0.6–0.9, 50 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the ≥ 0.9 band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in `report.ink.json`.